WORKED SOLUTION

STEP II 2002, Question 12

Probability and Statistics • Normal and Poisson Approximations

STEP II 2002, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q12

The key idea

There are three layers of randomness stacked inside each other: heads within a coin, coins within a day, and days within the week. Each layer is a binomial count whose "success" probability comes from the layer below. The formula is built by approximating the bottom layer with a normal distribution (with a continuity correction), the middle layer with a Poisson distribution and then a normal one, and leaving the top layer as an exact binomial. Once you see the three layers, the job is to identify which approximation produced each expression in the formula, and then to ask whether each approximation is any good for the given numbers.

Solution

One coin on one day

Fix a day and a coin. Let X be the number of heads in its M throws. The throws are independent, each with probability p of a head, so

$$X \sim \mathrm{B}(M, p), \qquad \mathrm{E}(X) = Mp, \qquad \mathrm{Var}(X) = Mp(1-p).$$

For large M we approximate X by a normal variable $X' \sim \mathrm{N}(Mp,\, Mp(1-p))$. Since X is integer-valued, "fewer than m heads" means $X \leq m-1$, and with a continuity correction this becomes $X' < m - \frac{1}{2}$. So

$$r = \mathrm{P}(X < m) = \mathrm{P}(X \leq m-1) \approx \Phi\left(\frac{m - \frac{1}{2} - Mp}{\sqrt{Mp(1-p)}}\right) = \Phi\left(\frac{2m - 1 - 2Mp}{2\sqrt{Mp(1-p)}}\right).$$

Comparing with the question, this says $h \approx Lr$.

All the coins on one day

Let Y be the number of the L coins that produce fewer than m heads on that day. The coins are thrown independently and each behaves in the same way, so each coin

independently "succeeds" with probability r, and

$$Y \sim \mathrm{B}(L, r), \qquad \mathrm{E}(Y) = Lr = h, \qquad \mathrm{Var}(Y) = Lr(1-r) = h(1-r).$$

Ask yourself The formula for q has $\sqrt{h}$ in the denominator. Why is the standard deviation $\sqrt{h}$ and not $\sqrt{h(1-r)}$?

The formula is designed for the case where r is small: fewer than m heads is a rare event for a single coin, while L is large. In that regime $1 - r \approx 1$, so the variance is approximately the mean, which is exactly the situation of the Poisson approximation $Y \approx \mathrm{Po}(h)$. If h is also reasonably large, $\mathrm{Po}(h)$ is in turn approximately $\mathrm{N}(h, h)$. (Equivalently, approximate $\mathrm{B}(L, r)$ directly by a normal and replace $1 - r$ by 1.)

So take $Y' \sim \mathrm{N}(h, h)$. "More than l coins" means $Y \geq l + 1$, which with a continuity correction becomes $Y' > l + \frac{1}{2}$. Hence

$$q = \mathrm{P}(Y > l) \approx 1 - \Phi\left(\frac{l + \frac{1}{2} - h}{\sqrt{h}}\right) = \Phi\left(\frac{h - l - \frac{1}{2}}{\sqrt{h}}\right) = \Phi\left(\frac{2h - 2l - 1}{2\sqrt{h}}\right),$$

using the symmetry $1 - \Phi(z) = \Phi(-z)$ of the standard normal distribution. This is the given q.

All the days

Call a day *good* if more than l of its coins produce fewer than m heads. Each day has the same probability q of being good, and different days involve different throws, so the days are independent. The number of good days among the K is therefore $\mathrm{B}(K, q)$, and

$$\mathrm{P}(\text{exactly } k \text{ good days}) = \binom{K}{k} q^k (1-q)^{K-k},$$

with q approximated as above. That is the required result.

The numerical case

Here $Mp = 60$ and $Mp(1-p) = 24$, so

$$\frac{2m - 1 - 2Mp}{2\sqrt{Mp(1-p)}} = \frac{96 - 1 - 120}{2\sqrt{24}} = \frac{-25}{9.798} \approx -2.55,$$

giving $r \approx \Phi(-2.55) \approx 0.0054$ and $h = 500r \approx 2.68$. Then

$$\frac{2h - 2l - 1}{2\sqrt{h}} \approx \frac{5.36 - 9}{3.27} \approx -1.11, \qquad q \approx \Phi(-1.11) \approx 0.133,$$

and the approximation gives $\binom{7}{2} q^2 (1-q)^5 \approx 21 \times 0.0178 \times 0.490 \approx 0.18$.

Ask yourself Each approximation is standard. Which of them are we using outside the conditions where it is reliable?

Take the three layers in turn.

- **Top layer.** The $\mathrm{B}(7, q)$ step is exact (given q), so it introduces no error of its own. But it squares the error in q, as $k = 2$.
- **Bottom layer.** $M = 100$ with $Mp = 60$ and $M(1-p) = 40$ is comfortably in the range where the normal approximation to the binomial is usually used. But we are using it 2.55 standard deviations into the tail, to find a probability of about 0.005. The normal approximation is good for the bulk of the distribution in *absolute* terms; in the tails its *relative* error can be large, especially as $p \neq \frac{1}{2}$ makes the binomial skewed. Every relative error in r passes straight into $h = Lr$, and q is sensitive to h.
- **Middle layer.** Replacing $\mathrm{B}(500, r)$ by $\mathrm{Po}(h)$ is excellent, since r is tiny and L large. But then approximating $\mathrm{Po}(h)$ by a normal with $h \approx 2.7$ is not justified: a Poisson distribution with mean under 3 is markedly skewed, and the usual rule of thumb asks for a mean of at least about 10 (some say 15 or 20).

So no, we would not expect the approximation to be accurate. The doubtful steps are the normal approximation to the tail of $\mathrm{B}(100, 0.6)$ and the normal approximation to a Poisson distribution with small mean.

In fact the exact value of r is $\mathrm{P}(X \leq 47) \approx 0.00576$, about 7% more than the normal estimate, so h should be about 2.88; and then the exact $\mathrm{P}(Y > 4)$ is about 0.165, not 0.133. The final probability is about 0.23 rather than 0.18. (For these particular numbers the normal approximation to $\mathrm{Po}(2.68)$ happens to give almost the same tail as the Poisson itself, 0.133 against 0.134, so most of the damage is done in the bottom layer. You could not have counted on that luck in advance.)

Checks

- The continuity corrections point the right way: "fewer than m" is $X \leq m - 1$, so we stop at $m - \frac{1}{2}$; "more than l" is $Y \geq l + 1$, so we start at $l + \frac{1}{2}$. The given formulae contain exactly $2m - 1$ and $2l + 1$.
- Numerically, $\Phi(-2.5516) = 0.005362$, $h = 2.681$ and $q = \Phi(-1.1108) = 0.1333$, giving 0.1825. Exact binomial sums give $r = 0.005761$, $q = 0.1645$ (confirmed by a simulation of 2×10^4 days, which gave 0.164) and a final answer of 0.2314.

Common slips

- Using m instead of $m - \frac{1}{2}$ (or l instead of $l + \frac{1}{2}$): then the formulae don't match.
- Using $\sqrt{h(1-r)}$ and then being unable to reach $\sqrt{h}$. The question's formula drops the factor $1 - r$ because it is effectively using the Poisson approximation, and you should say so.

- Judging accuracy only by "$Mp > 5$ and $M(1-p) > 5$". That rule is about the centre of the distribution, not about a tail probability of 0.005.

Beyond STEP

The question is about how errors propagate through a chain of approximations, and about the difference between absolute and relative error in the tails. At university, the central limit theorem is sharpened by the Berry–Esseen theorem, which bounds the absolute error of the normal approximation. Tail probabilities need different tools, such as Chernoff bounds and large-deviation theory, which describe how fast tail probabilities shrink. Blitzstein and Hwang, *Introduction to Probability*, covers the Poisson and normal approximations and Chernoff-type inequalities clearly; Grimmett and Welsh, *Probability: An Introduction*, is a good companion.

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